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To:
Editor-in-Chief, IEEE Transactions on Information Forensics and Security (TIFS)
IEEE Signal Processing Society

Subject: Submission of Original Research Manuscript: “C-STGB: Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering under Extreme Imbalance and Topological Camouflage”

Dear Editor-in-Chief and Editorial Board,

We are pleased to submit our original research manuscript titled “C-STGB: Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering under Extreme Imbalance and Topological Camouflage” for consideration for publication in IEEE Transactions on Information Forensics and Security (TIFS).

Research Context & Problem Formulation: Anti-Money Laundering (AML) surveillance represents a critical application of graph learning in financial forensics. While Graph Neural Networks (GNNs) capture relational transaction topologies, real-world deployment faces five foundational challenges: (1) continuous-time velocity evasion across micro-bursts and multi-month dormant periods; (2) adversarial topological camouflage via high-degree commercial hubs; (3) extreme class imbalance (illicit prevalence below 0.05%) causing severe recall collapse; (4) cold-start graph isolation for disconnected entities; and (5) uncalibrated decision uncertainty.

Methodological Innovations: To resolve these interconnected bottlenecks, we introduce C-STGB (Conformal Spatio-Temporal GraphBoost), a unified geometric framework featuring:

1. Tri-Band Continuous Harmonic Attention: Disentangles microsecond smurfing bursts from 90-day dormant holding chains via continuous temporal encodings and localized Hawkes arrival intensities.
2. Context-Aware Edge Trust Gating ($\hat{g}_{ij}$): Suppresses spurious camouflage linkages to high-degree commercial hubs, mitigating feature over-smoothing.
3. Typology-Clustered Latent GraphSMOTE: Synthesizes virtual minority entities along latent manifolds with bilinear edge generation, recovering minority recall without precision collapse.
4. Evidence-Adaptive Graph-Tabular Fusion: Dynamically transfers representation weight to 12-D canonical flow invariants for cold-start entities ($\deg(u) = 0$).
5. Class-Conditional Conformal Risk Control (CRC): Provides finite-sample coverage guarantees ($1 - \alpha \geq 99.0\%$) under stated within-class exchangeability assumptions and routes over 99.4% of volume through automated clearing/escalation tiers.
6. Multi-Agent Compliance Verification: Generates audit-traceable FinCEN Form 111 XML dossiers with formal schema validation preventing unsupported entity references.

Empirical Evaluation & Findings: Evaluated across 13 diverse financial transaction networks encompassing over 8.81 million entities and 11.42 million transactions across five random seeds, our proposed C-STGB achieves the highest performance among evaluated baselines across the benchmarks ($91.42 \pm 0.65\%$ F1 on Elliptic-v1, $86.54 \pm 0.82\%$ F1 on Elliptic-v2, and $94.62 \pm 0.55\%$ F1 on Ethereum Phishing). In our streaming inference evaluation, C-STGB achieves a single-event latency of 0.45 ms (Fast-Path) and 2.10 ms (Full GNN), well below the 10 ms target latency.

Declarations & Supplementary Package: The submission package comprises our main manuscript and an accompanying Supplementary Material document containing formal algorithmic specifications, extended mathematical proofs, and comprehensive reproducibility configurations. We confirm that this manuscript represents our original work that has not been published and is not under consideration elsewhere. All authors have reviewed and approved the submission. We declare no competing financial or personal conflicts of interest. The complete open-source codebase and reproducibility artifacts are maintained at: <https://github.com/NazmulHasanNihal/Intelligent-AML>.

Sincerely yours,

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